

2023



Abit Agritech Builds a Minimum Viable Product in 6 Months Using AWS Amplify

Learn how Abit Agritech, an agricultural software startup, created a mobile app minimum viable product in 6 months using AWS Amplify.

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6 months

development time for a minimum viable product | in costs compared to an on-premises solution

68% improved

time to market

Empo

with data-driven sustainability insights



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Overview

Agricultural startup [Abit Agritech](#) (Abit) created the minimum viable product (MVP) of its mobile app in 6 months using Amazon Web Services (AWS) while facing a tight deadline to create a flexible application with scalable features. The company's focus is on biodiversity and sustainable farming. To improve the technology and tangibility of data for farmers, Abit decided to build a mobile app to provide farmers with data-driven insights about soil quality, weather patterns, biodiversity, and more to support sustainable agricultural practices.

Aiming to quickly develop an MVP to be ready in time for the farming season, the company used [AWS Amplify](#), a complete solution that front-end web and mobile developers can use to easily build, ship, and host full-stack applications on AWS. The company uses AWS Amplify Studio, Amplify's visual interface, for its frontend UI. Abit has subsequently improved development speed, improved time to market, and reduced costs.



Opportunity | Using AWS Amplify to Improve Development Speed for a New Abit Agritech Mobile App

Founded in 2022, the company of five people based in Milan, Italy, emerged around a shared mission to build an agricultural technology company focused on sustainable solutions. Abit focuses on improving biodiversity, in response to the European Union's Biodiversity Strategy for 2030, by providing technology and data to farmers and supporting sustainable agriculture. As a startup, Abit knew from the beginning that it wanted the flexibility and scalability offered by serverless cloud architecture so that it would have room to grow. "We chose AWS because we needed to deploy a scalable solution with our business going through a series of massive changes as a startup," says Chiara Antonucci, CEO and cofounder of Abit. The company reached out to AWS with questions about serverless solutions. Within 1 month, Abit had chosen to work with [Corley srl](#) (Corley), an [AWS Partner](#), for their expertise.



The MVP mobile app for iOS and Android is designed to gather data about climate and soil conditions using GPS and then provide this information to farmers in a tangible way. Along with this comes a certification for the biodiversity index that can be recognized by stakeholders and gives farmers simplified access to public incentives, like those in the Biodiversity Strategy for 2030. The app also provides farmers with dedicated support from the Abit team. To create the MVP app and deploy it quickly, the company used AWS Amplify. "AWS Amplify was important to us because it works like a glue for AWS services," says Antonucci. "It was much easier for us to put together various services for the app." Abit benefited from having everything in one place, using AWS Amplify Studio to create a rich UI and connect the UI to a backend in a matter of clicks. Working with Corley and using AWS Amplify, Abit developed its MVP app in 6 months, a process that would have taken twice as long otherwise.

Solution | Saving Costs While Improving Farmers' Sustainability

Speed was essential for the release of Abit's MVP because Abit wanted to align the release of its app with the peak of the farming season. Using AWS Amplify, the company increased its speed to market by 68 percent. This boost was possible because of improved agility using AWS services, which in turn increased developer speed. "Using AWS Amplify, we developed faster with two developers," says Antonucci. Abit sped up front-end developers by using AWS Amplify to transform its interface design application into code usable for the MVP app. Because Abit uses AWS Amplify to handle the heavy lifting for code, its developers can focus on innovating rather than managing backend operations.

Abit also uses AWS Amplify to incorporate other AWS services into its MVP app. For data collection, Abit uses [Amazon DocumentDB \(with MongoDB compatibility\)](#), a fully managed native JSON document database that makes it simple and cost effective to operate critical document workloads at virtually any scale without managing infrastructure. "Amazon DocumentDB is helpful for us because we use it to store all the data that we have, including climate and temperature data," says Antonucci.

By choosing serverless solutions using AWS, Abit saves on costs because it doesn't need to host or manage servers. Compared to a local or on-premises solution, Abit has a 70 percent reduction in costs using serverless solutions on AWS. "We turn this cost reduction directly back to our customers so that they can benefit from a smart solution at a cheaper price," says Antonucci. Additionally, Abit's serverless architecture gives the company the opportunity to test in new marketplaces with an increase only in operating costs. "Using AWS services, we can grow our marketplace with no increase in fixed costs," says Antonucci. "This is very important to us as a growing and entering a new marketplace to see if it fits with our MVP application."



Most importantly, Abit's MVP app works well for farmers promoting biodiversity and benefits the end customer. Farmers can use the app to register their fields and, thanks to GPS localization, can see relevant climate information, such as temperature and condensation, to improve everyday field management. Abit can calculate the biodiversity index for the field, and the farmer can access dedicated support provided by the Abit team by using the app. By using these weather and soil insights, farmers are better equipped to prepare crops with sustainability and biodiversity in mind. "The MVP app meets our expectations, and our customers are very satisfied," says Antonucci. "The app helps farmers make the right decisions for everyday management of their fields."

Outcome | Continuing to Develop and Innovate

Abit plans to introduce the next version of its app in February 2024. Creating the MVP in 6 months helped set up the business for success. Abit intends to continue improving its mobile app using the extensibility capabilities of AWS Amplify and AWS Amplify Studio to add richer app functionality and to scale to more features for more farmers. "Using AWS services and a partner like Corley, we built an MVP and first release with only two developers, and that was very beneficial for us," says Antonucci.

About Abit Agritech

Abit Agritech uses innovative technology to suggest more efficient crop management, with the goal of protecting biodiversity, and provides farmers with the knowledge that they need to improve sustainable agriculture.

AWS Services Used

AWS Amplify

AWS Amplify lets frontend web and mobile developers easily build, ship, and host full-stack applications on AWS, with the flexibility to leverage the breadth of AWS services as use cases evolve. No cloud expertise needed.

[Learn more »](#)

Amazon DocumentDB

Amazon DocumentDB (with MongoDB compatibility) is a fully managed native JSON document database that makes it easy and cost effective to operate critical document workloads at virtually a



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